Car Price Prediction

Using Machine Learning to Predict Automobile Prices

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MSIT — 2026

Models: Linear Regression | Decision Tree | XGBoost

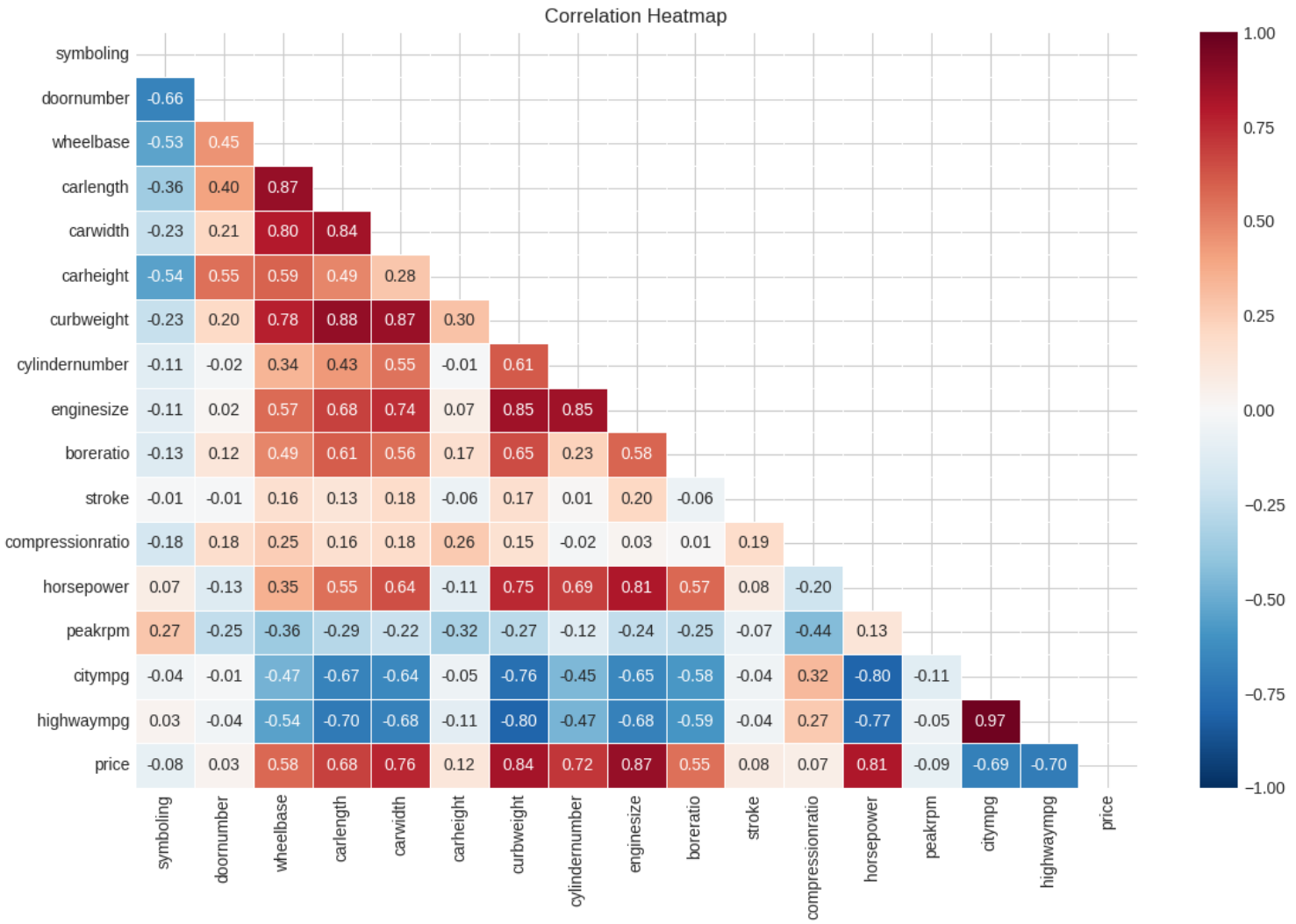
Dataset: CarPrice Assignment (205 cars, 26 features)

Tools: Python, Scikit-learn, XGBoost, SHAP, LIME

Feature Engineering

Feature Engineering:

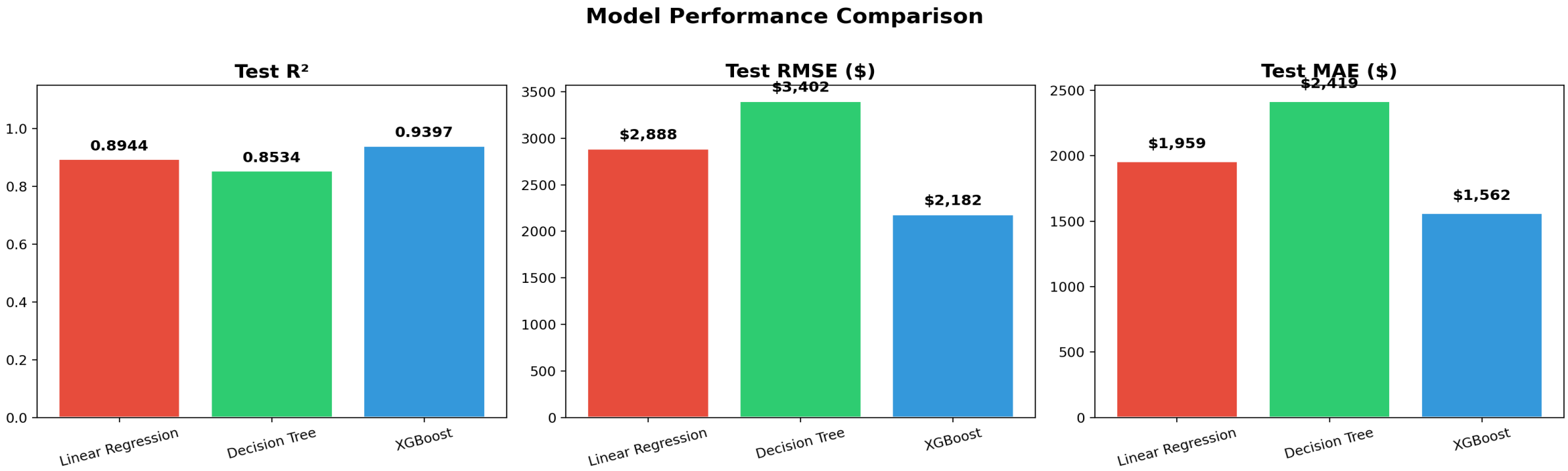
- Dropped highway mpg — almost identical to city mpg (0.97 correlation)
- Combined horsepower and weight into a **power_to_weight** ratio
- Extracted 22 brands from the car name field
- Grouped brands into **brand_tier**: luxury / mid-range / economy
- Created log transforms for skewed features (price, engine size)
- Dropped low-signal features (stroke, compression ratio, peak rpm)



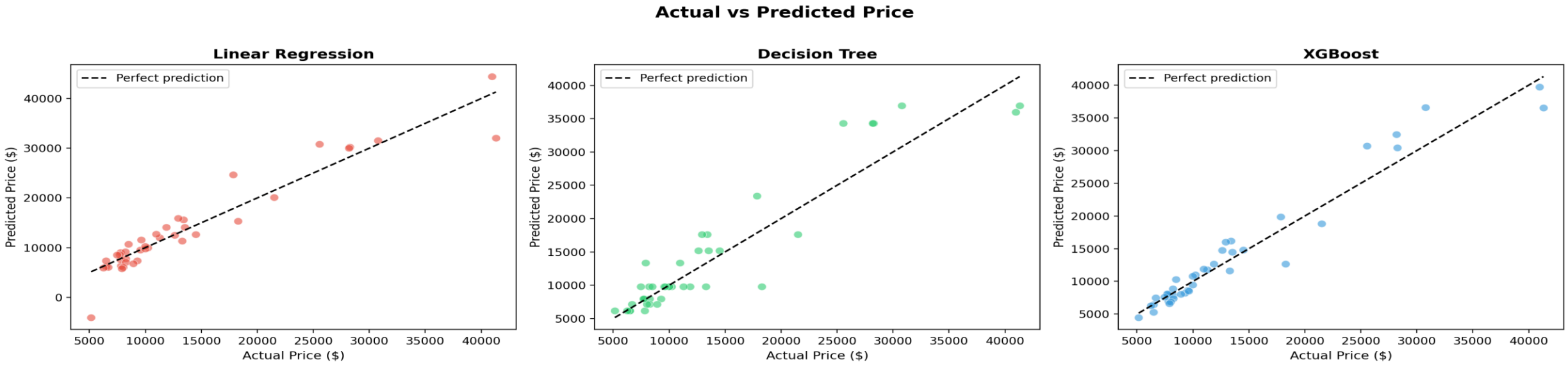
Phase 1: Manual Model Performance Comparison

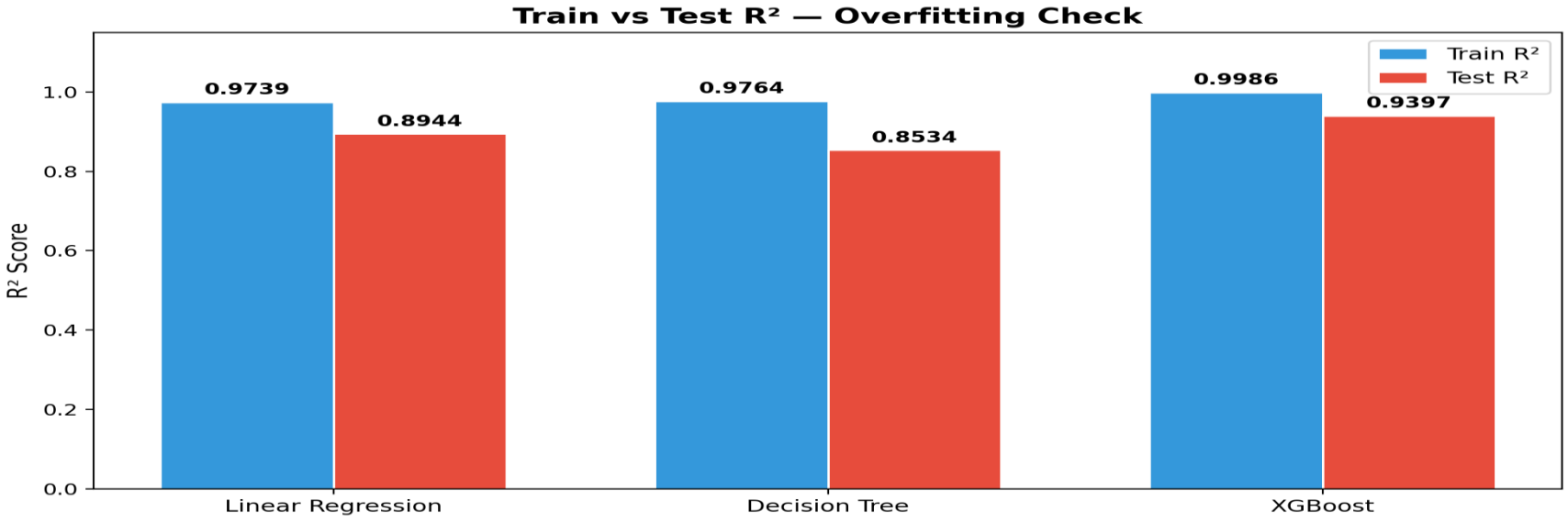
Model Performance Summary

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	Train MAE	Test MAE
Linear Regression	0.9739	0.8944	$1,247	$2,888	$956	$1,959
Decision Tree	0.9764	0.8534	$1,186	$3,402	$875	$2,419
XGBoost	0.9986	0.9397	$288	$2,182	$103	$1,562

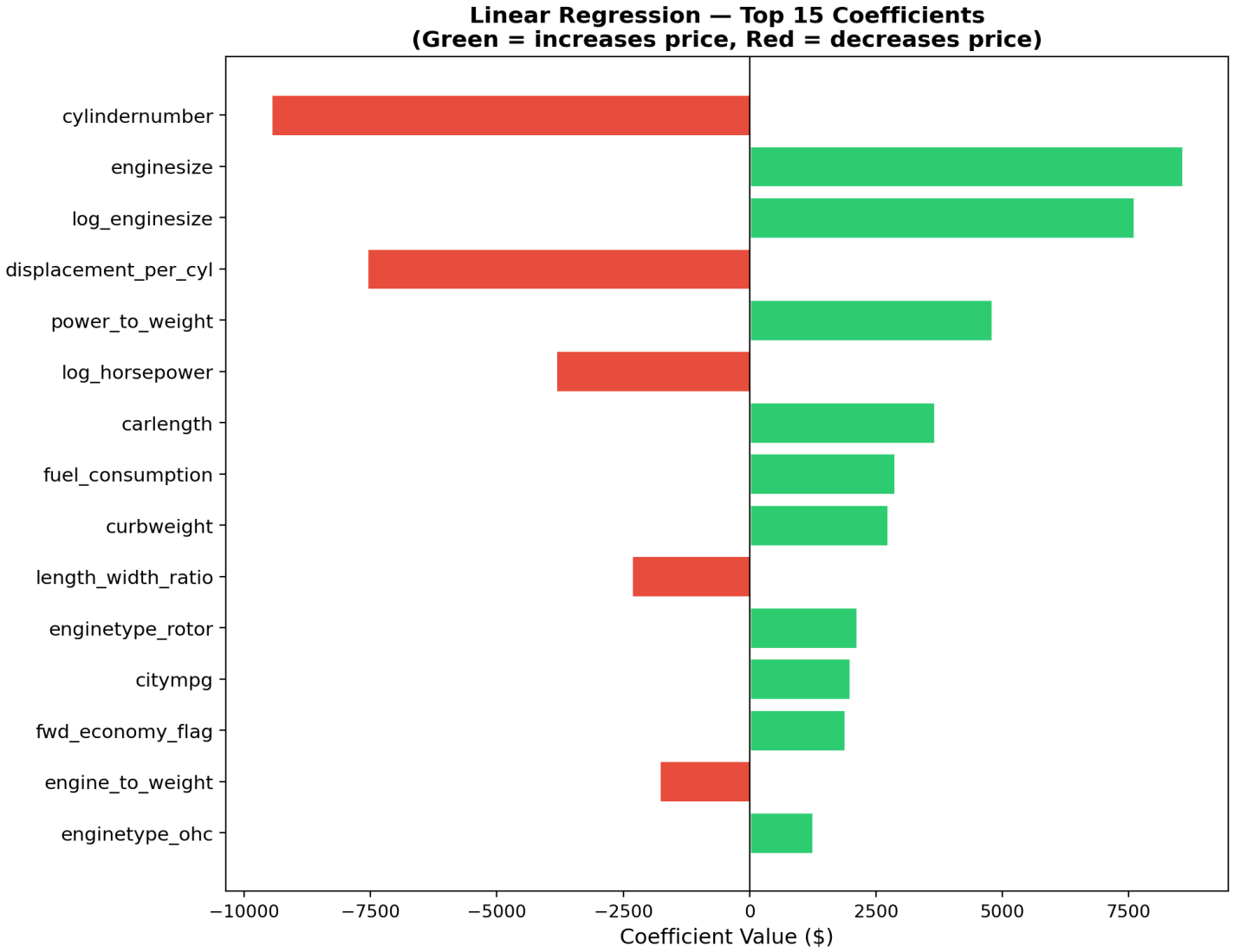


Prediction Accuracy & Overfitting Check





Linear Regression — Coefficient Interpretation



Key Interpretations:

Dummy Variable:

enginetype_rotor

increases price by $2,120

vs the baseline category

Quantitative Variable:

enginesize

+1 unit increases price

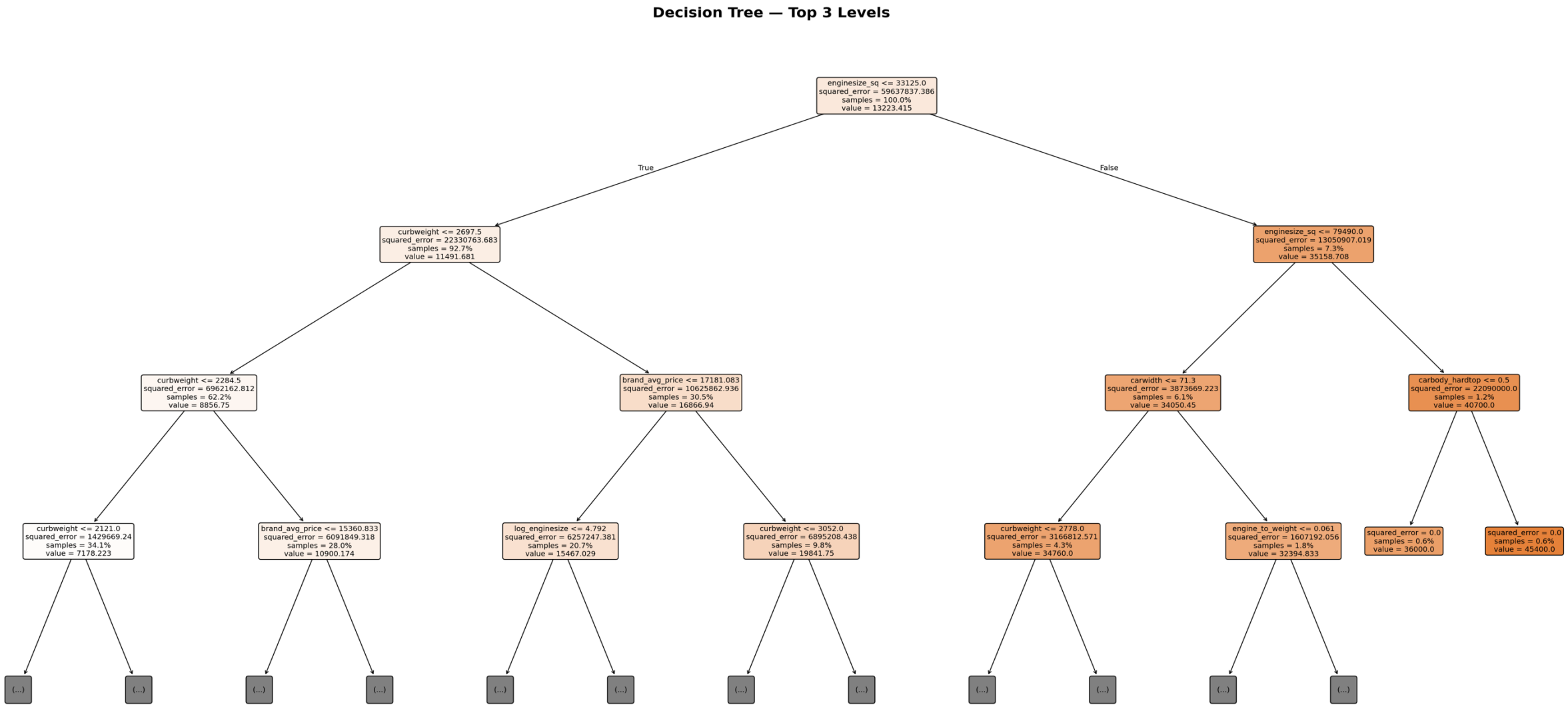
by $8,569

Limitation:

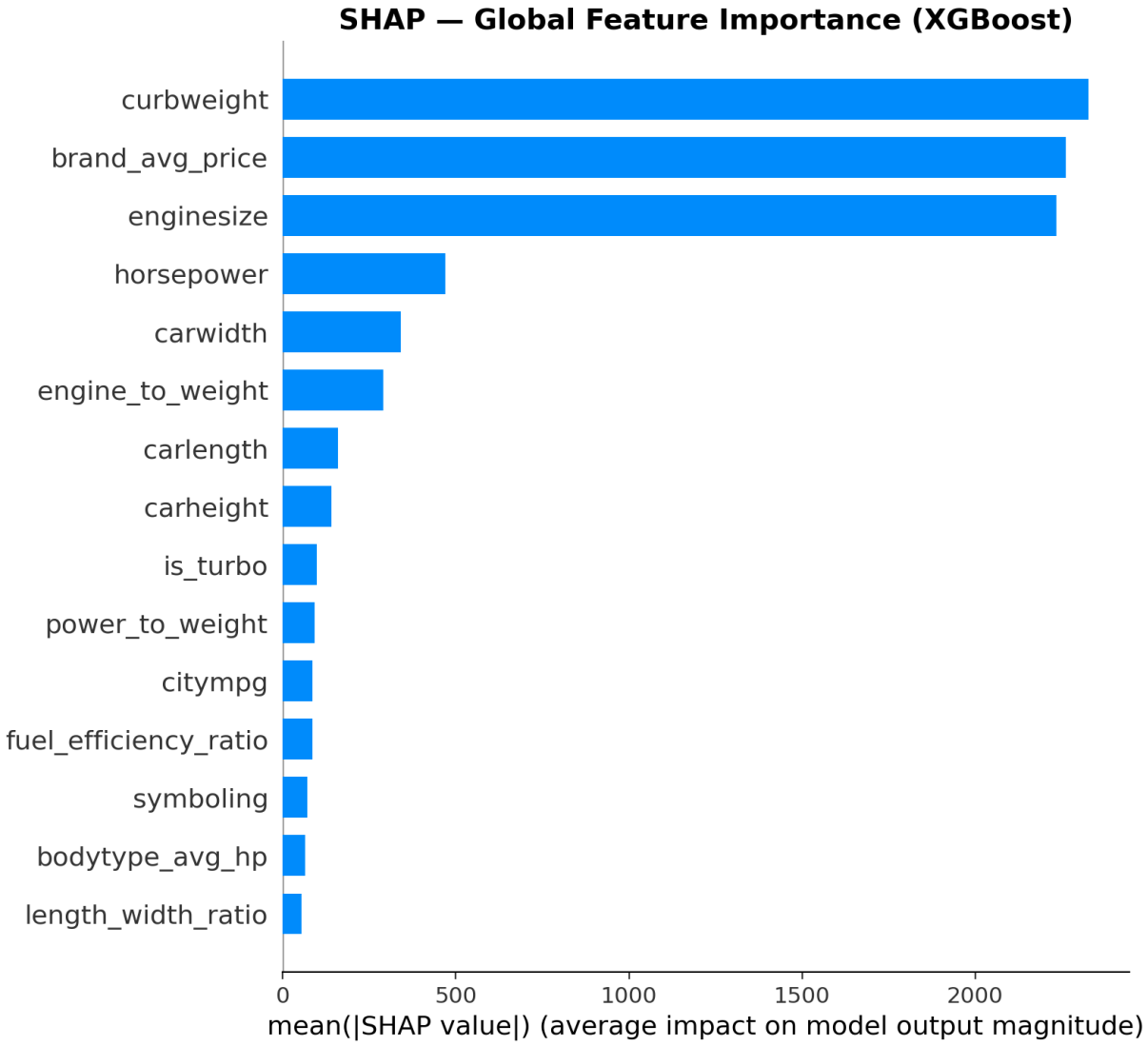
Coefficients may be unreliable

if data is skewed

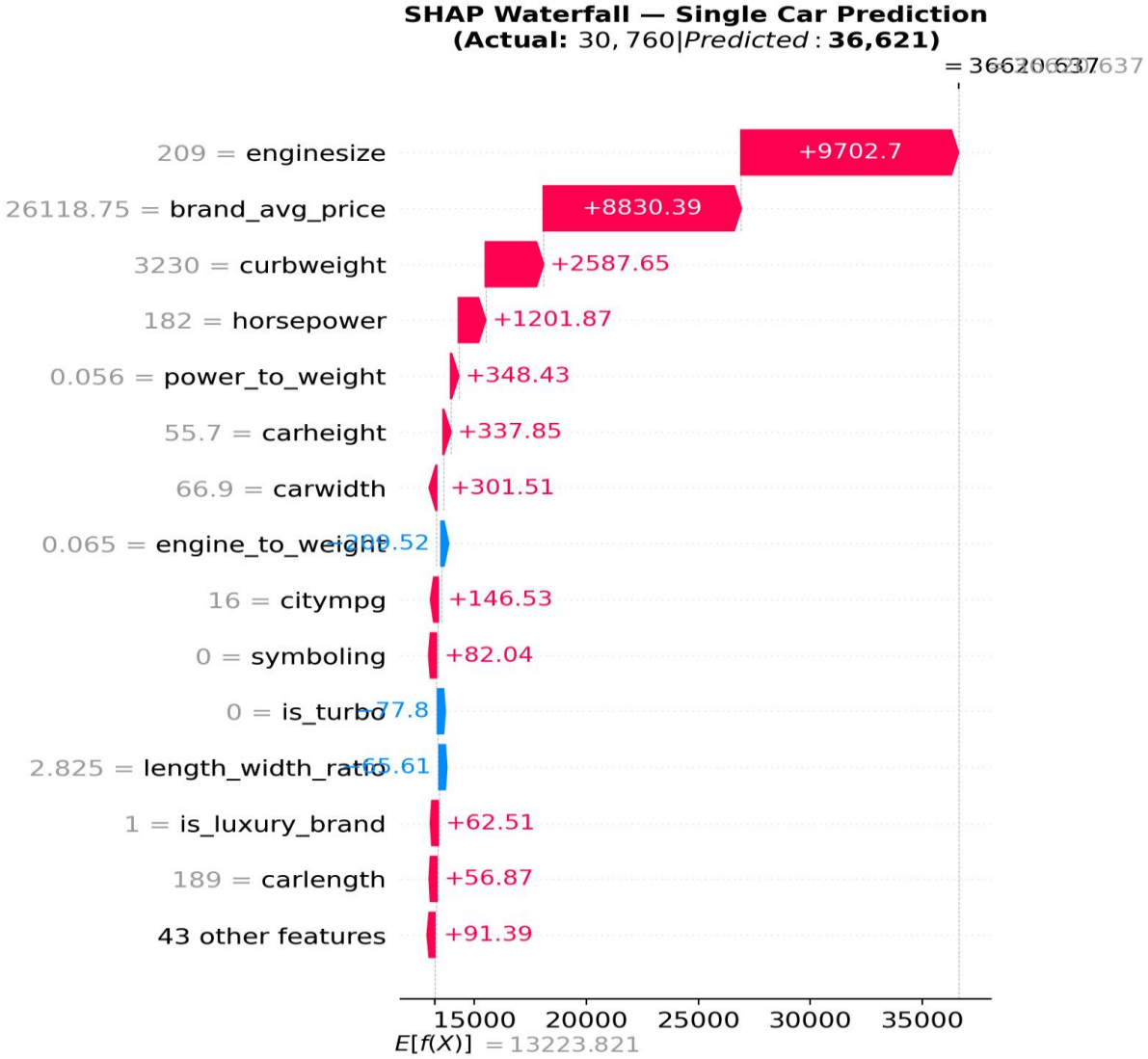
Decision Tree — Node Interpretation



SHAP — Global & Local Explainability (XGBoost)

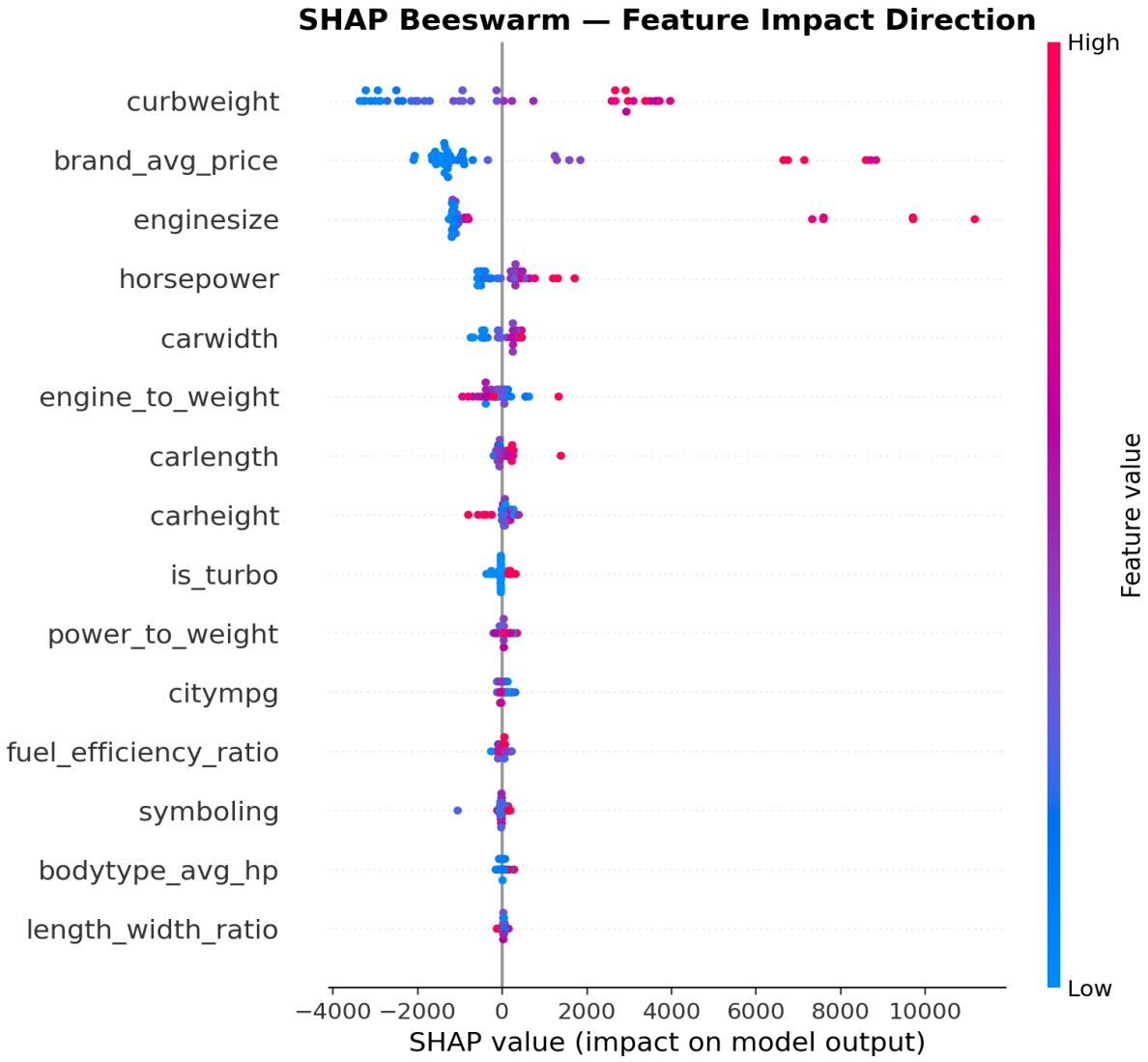


Global: Which features matter most?

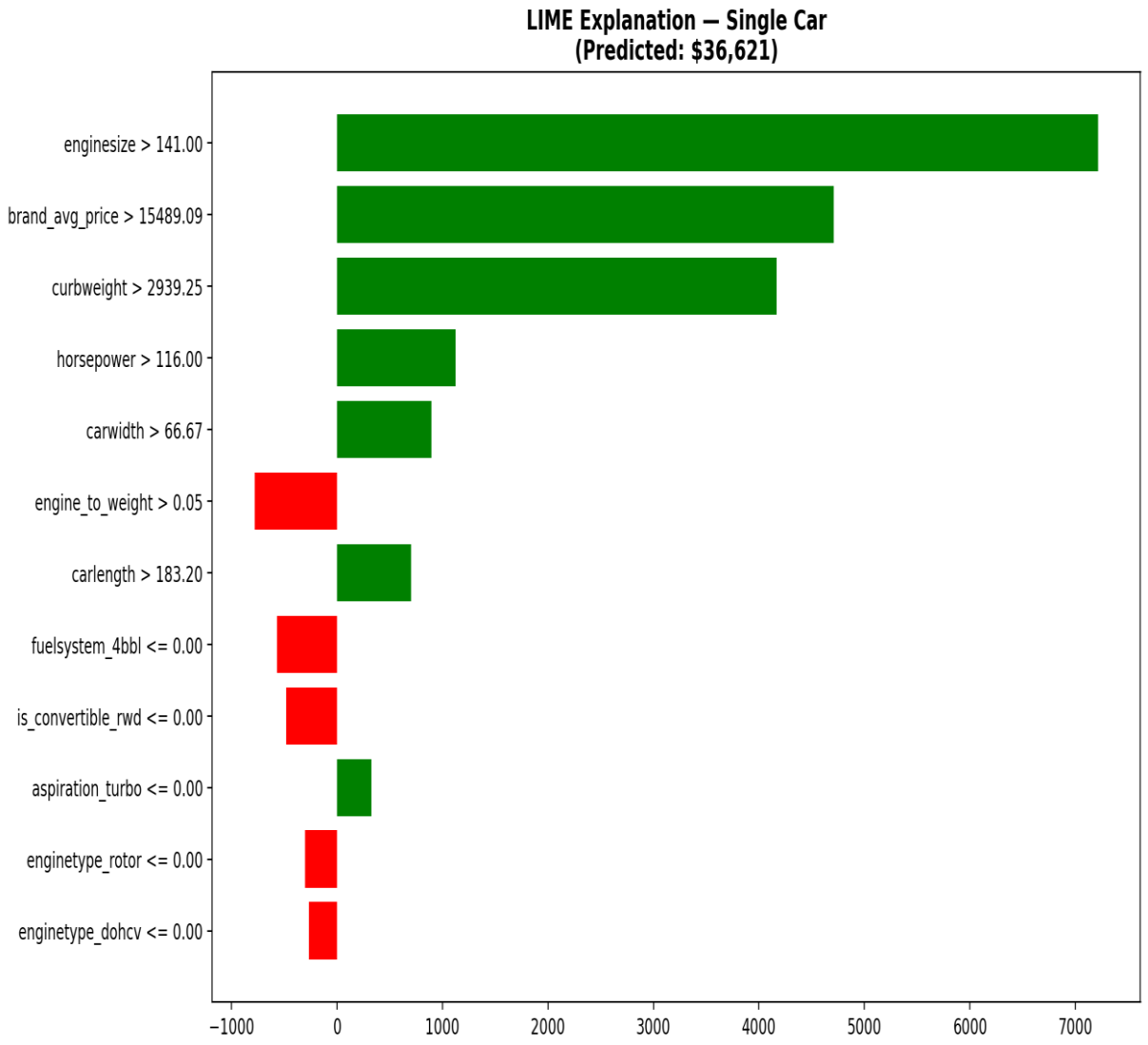


Local: Why this price for this car?

SHAP Beeswarm & LIME Explanation



SHAP Beeswarm: Impact direction per feature



LIME: Local model explanation

PyCaret Comparison – Automated Machine Learning Benchmark

MODEL COMPARISON TABLE

Sorted by RMSE (Best First)

Rank	Model	Train R^2	Test R^2	Train RMSE	Test RMSE	Train MAE	Test MAE
1	**Extra Trees**	**0.998**	**0.949**	**$332**	**$2,008**	**$154**	**$1,436**
2	**Random Forest**	0.987	0.947	$876	$2,046	$557	$1,426
3	**Bagging**	0.988	0.944	$848	$2,095	$537	$1,453
4	**CatBoost**	0.995	0.944	$560	$2,102	$426	$1,448
5	**XGBoost**	0.999	0.940	$288	$2,182	$103	$1,562
6	**LightGBM**	0.984	0.927	$963	$2,394	$562	$1,734
7	**Gradient Boosting**	0.999	0.919	$284	$2,523	$79	$1,704
8	**AdaBoost**	0.965	0.909	$1,435	$2,683	$1,219	$2,039
9	**Huber Regressor**	0.969	0.899	$1,367	$2,821	$926	$1,878
10	**Linear Regression**	0.974	0.894	$1,247	$2,888	$956	$1,959
11	**Lasso**	0.974	0.893	$1,251	$2,903	$962	$1,972
12	**ElasticNet**	0.939	0.881	$1,905	$3,061	$1,369	$2,027
13	**Ridge**	0.969	0.880	$1,368	$3,080	$1,032	$2,022
14	**Bayesian Ridge**	0.960	0.873	$1,537	$3,170	$1,111	$2,161
15	**Decision Tree**	0.974	0.853	$1,186	$3,402	$875	$2,419
16	**KNN**	0.899	0.844	$2,456	$3,513	$1,570	$2,353
17	**SVR**	0.130	0.096	$7,204	$8,449	$4,332	$4,704

Green = stronger performance | Red = weaker performance

- PyCaret tested 17 regression models automatically
- Used cross-validation for ranking
- Best automated model = Extra Trees
- Manual XGBoost remained highly competitive
- Difference between top models was small

Conclusion & Key Takeaways

Model Performance

Best Manual Model: XGBoost

Test R^2: 0.9397 (94.0% variance explained)

Test RMSE: $2,182

RMSE as % of avg price: 16.2%

Progression:

LR: $2,888 RMSE

DT: $3,402 RMSE

XGB: $2,182 RMSE

Key Findings:

- Engine size, curb weight, brand tier strongest predictors
- Feature engineering improved performance
- Explainability tools validated predictions

Limitations:

- Small dataste (205 rows)
- No mileage / year / inflation variables

Next Steps:

- Larger dataset
- Hyperparameter tuning
- Real market deployment

Thank You!

Questions & Discussion

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Car Price Prediction Project